

Parts checklist

Tripoli air quality monitoring: one sensor node and one gateway. Status as of 7 October 2026.

Where things stand

Working today: the gateway is online, and the sensor board joins the network and sends a message every 5 minutes, which the server receives and decodes.

Not working yet: real readings. Neither sensor can be connected until the three items under "Buy first" are sorted.

Buy first

Item	What to search for	Why
PMS7003 adapter board with cable	"PMS7003 adapter" or "G7 adapter" (1.27 mm to 2.54 mm)	Converts the dust sensor's tiny connector to normal pins
Jumper wires	"Dupont jumper wires", mixed pack with female-female and male-female	Connects the sensors to the board
Soldering	A basic soldering iron and solder, or a few minutes at a phone repair shop	Attaches the loose pin strips (see "What needs soldering")

Sensor node

Item	Status	Notes
T-Beam V1.2 board, 868 MHz, with LoRa antenna	Have	Label Q401+L206-MESH. Radio chip SX1276, power chip AXP2101. Tested and working.
PMS7003 dust sensor	Have	Cannot be connected until the adapter arrives.
BME280 temperature and humidity sensor	Have	Its 4-pin strip is loose and needs soldering on.
TP4056 charger (USB-C, with protection)	Set aside	Probably not needed: the T-Beam has its own battery charger. Never connect two chargers to one battery.
USB data cable	To confirm	A borrowed one works. Get one to keep with the kit. It must be a data cable, not charge-only.
PMS7003 adapter board with cable	Missing	See "Buy first".
Jumper wires	Missing	See "Buy first".
Soldering iron and solder, or a shop	Missing	See "Buy first".
18650 battery	To confirm	The parts list names the Samsung INR18650-30Q. Insert it the right way round: reversed polarity can destroy the board.
5 V solar panel, 2 W, USB plug	To confirm	Plugs into the board's USB port.
Waterproof box and cable glands	To confirm	See design points 2 and 3 before buying more.

Item	Status	Notes
5 V step-up (booster) module, 3.7 V to 5 V	Likely missing	Not on the original parts list. See design point 1.

Gateway

Item	Status	Notes
Dragino DLOS8N-EC25 with LoRa antenna and extension	Have	Gateway ID a84041ffff2e6ba0, serial os8lp505679. Online over Wi-Fi.
Power adapter	Have	To switch the gateway off, unplug it. Settings are kept.
Two bendable metal strips	Have	Almost certainly the pole-mounting straps. Only needed when the gateway goes outdoors.
4G antenna	To confirm	Check the gateway for a second, empty antenna connector. If there is none, the antenna is built in.
SIM card with a data plan	Missing	Only needed if the gateway will run on 4G in the field.
Backup power	Missing	The gateway must stay powered around the clock. With daily outages it needs a UPS, or its own battery and solar setup. It draws roughly 5 W.
Outdoor power or network cable	Missing	Long enough to reach the mounting spot.
Pole or mast, placed high	Missing	Height matters more than anything else for range.

What needs soldering

- The 4-pin strip onto the BME280.
- The pins onto the PMS7003 adapter, if they arrive loose.
- The T-Beam's own pin strips, if its edges have only empty holes. This has not been checked yet.

If a shop does the work, take all of these in one visit.

Three design points to settle before boxing a node

These have not been tested on the hardware yet. They are the first things to check once the sensors are wired.

1. 5 V for the dust sensor. The PMS7003 needs 5 V. The T-Beam's 5 V pin is fed from USB, so on battery alone the sensor would likely get no power. A small step-up module fixes that.

2. Air has to reach the sensors. A fully sealed box blocks the dust sensor's airflow, and a temperature sensor inside a sun-heated box reads the box, not the air. The enclosure needs downward-facing vents, or the sensors in a shaded, vented section.

3. The box size. The listed box is 25 mm deep. The T-Beam with a battery in its holder, plus the dust sensor, may not fit. Measure before ordering more.

Wiring reference

For when the parts arrive. Unplug the USB cable before wiring.

Sensor	Sensor pin	T-Beam pin
BME280	VCC (or VIN)	3V3 (not 5V: 5 V can damage it)
BME280	GND	GND
BME280	SDA	21
BME280	SCL	22
PMS7003	VCC	5V
PMS7003	GND	GND
PMS7003	TX	13
PMS7003	RX	14